ASSIGNMENT-2

Database: `library_db`

Collections:

* `members`

* `books`

* `borrowings`

➔ use library_db

switched to db library_db

➔ db.createCollection("members")

{ ok: 1 }

➔ db.createCollection("books")

{ ok: 1 }

➔ db.createCollection("borrowings")

{ ok: 1 }

➢ DATA INSERTION:

```
db.members.insertMany([
 { member_id: 1, name: "Aarav", city: "Hyderabad", membership_type: "Gold" },
 { member_id: 2, name: "Priya", city: "Bangalore", membership_type: "Silver" },
 { member_id: 3, name: "Rahul", city: "Mumbai", membership_type: "Gold" },
 { member_id: 4, name: "Sneha", city: "Delhi", membership_type: "Silver" },
 { member_id: 5, name: "Kiran", city: "Hyderabad", membership_type: "Gold" }
])
```

o/p:

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69df08bf5b823e6a174a704a'),
    '1': ObjectId('69df08bf5b823e6a174a704b'),
    '2': ObjectId('69df08bf5b823e6a174a704c'),
    '3': ObjectId('69df08bf5b823e6a174a704d'),
    '4': ObjectId('69df08bf5b823e6a174a704e')
  }
}
db.books.insertMany([
{ book_id: 101, title: "MongoDB Basics", category: "Database", author: "John Smith",
price: 500 },
```

```
 { book_id: 102, title: "Python Fundamentals", category: "Programming", author: "Alice Brown", price: 650 },
 { book_id: 103, title: "Data Engineering Intro", category: "Data", author: "Mark Lee", price: 800 },
 { book_id: 104, title: "SQL for Beginners", category: "Database", author: "David Miller", price: 450 },
 { book_id: 105, title: "Machine Learning Start", category: "AI", author: "Sara Khan", price: 900 }
])
```

o/p:

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69df08c95b823e6a174a704f'),
    '1': ObjectId('69df08c95b823e6a174a7050'),
    '2': ObjectId('69df08c95b823e6a174a7051'),
    '3': ObjectId('69df08c95b823e6a174a7052'),
    '4': ObjectId('69df08c95b823e6a174a7053')
  }
}
db.borrowings.insertMany([
 { borrow_id: 1001, member_id: 1, book_id: 101, days_borrowed: 5, borrow_date: "2026-03-01" },
 { borrow_id: 1002, member_id: 2, book_id: 102, days_borrowed: 3, borrow_date: "2026-03-02" },
 { borrow_id: 1003, member_id: 1, book_id: 103, days_borrowed: 7, borrow_date: "2026-03-03" },
 { borrow_id: 1004, member_id: 3, book_id: 104, days_borrowed: 4, borrow_date: "2026-03-05" },
 { borrow_id: 1005, member_id: 5, book_id: 105, days_borrowed: 10, borrow_date: "2026-03-07" },
 { borrow_id: 1006, member_id: 5, book_id: 101, days_borrowed: 2, borrow_date: "2026-03-08" }
])
```

o/p:

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69df08d55b823e6a174a7054'),
    '1': ObjectId('69df08d55b823e6a174a7055'),
    '2': ObjectId('69df08d55b823e6a174a7056'),
    '3': ObjectId('69df08d55b823e6a174a7057'),
    '4': ObjectId('69df08d55b823e6a174a7058'),
    '5': ObjectId('69df08d55b823e6a174a7059')
  }
```

}

Basic Queries

1. Display all members.

- > printjson(db.members.find())

o/p:

```
{
 _id: ObjectId('69df08bf5b823e6a174a704a'),
 member_id: 1,
 name: 'Aarav',
 city: 'Hyderabad',
 membership_type: 'Gold'
}
{
 _id: ObjectId('69df08bf5b823e6a174a704b'),
 member_id: 2,
 name: 'Priya',
 city: 'Bangalore',
 membership_type: 'Silver'
}
{
 _id: ObjectId('69df08bf5b823e6a174a704c'),
 member_id: 3,
 name: 'Rahul',
 city: 'Mumbai',
 membership_type: 'Gold'
}
{
 _id: ObjectId('69df08bf5b823e6a174a704d'),
 member_id: 4,
 name: 'Sneha',
 city: 'Delhi',
 membership_type: 'Silver'
}
{
 _id: ObjectId('69df08bf5b823e6a174a704e'),
 member_id: 5,
 name: 'Kiran',
 city: 'Hyderabad',
 membership_type: 'Gold'
}
```

---

2) Display all books.

printjson(db.books.find())

o/p:

```
{
  _id: ObjectId('69df08c95b823e6a174a704f'),
  book_id: 101,
  title: 'MongoDB Basics',
  category: 'Database',
  author: 'John Smith',
  price: 500
}
{
  _id: ObjectId('69df08c95b823e6a174a7050'),
  book_id: 102,
  title: 'Python Fundamentals',
  category: 'Programming',
  author: 'Alice Brown',
  price: 650
}
{
  _id: ObjectId('69df08c95b823e6a174a7051'),
  book_id: 103,
  title: 'Data Engineering Intro',
  category: 'Data',
  author: 'Mark Lee',
  price: 800
}
{
  _id: ObjectId('69df08c95b823e6a174a7052'),
  book_id: 104,
  title: 'SQL for Beginners',
  category: 'Database',
  author: 'David Miller',
  price: 450
}
{
  _id: ObjectId('69df08c95b823e6a174a7053'),
  book_id: 105,
  title: 'Machine Learning Start',
  category: 'AI',
  author: 'Sara Khan',
  price: 900
}
```

---

3. Display all borrowings.

printjson(db.borrowings.find())

o/p:

```
{
  _id: ObjectId('69df08d55b823e6a174a7054'),
  borrow_id: 1001,
  member_id: 1,
  book_id: 101,
  days_borrowed: 5,
  borrow_date: '2026-03-01'
}
{
  _id: ObjectId('69df08d55b823e6a174a7055'),
  borrow_id: 1002,
  member_id: 2,
  book_id: 102,
  days_borrowed: 3,
  borrow_date: '2026-03-02'
}
{
  _id: ObjectId('69df08d55b823e6a174a7056'),
  borrow_id: 1003,
  member_id: 1,
  book_id: 103,
  days_borrowed: 7,
  borrow_date: '2026-03-03'
}
{
  _id: ObjectId('69df08d55b823e6a174a7057'),
  borrow_id: 1004,
  member_id: 3,
  book_id: 104,
  days_borrowed: 4,
  borrow_date: '2026-03-05'
}
{
  _id: ObjectId('69df08d55b823e6a174a7058'),
  borrow_id: 1005,
  member_id: 5,
  book_id: 105,
  days_borrowed: 10,
  borrow_date: '2026-03-07'
}
{
  _id: ObjectId('69df08d55b823e6a174a7059'),
  borrow_id: 1006,
  member_id: 5,
```

```
  book_id: 101,
  days_borrowed: 2,
  borrow_date: '2026-03-08'
}
```

---

4. Show members from **Hyderabad**.

db.members.find({city:"Hyderabad"})

o/p:

```
{
  _id: ObjectId('69df08bf5b823e6a174a704a'),
  member_id: 1,
  name: 'Aarav',
  city: 'Hyderabad',
  membership_type: 'Gold'
}
{
  _id: ObjectId('69df08bf5b823e6a174a704e'),
  member_id: 5,
  name: 'Kiran',
  city: 'Hyderabad',
  membership_type: 'Gold'
}
```

---

5. Show books in the **Database** category.

printjson(db.books.find({category:"Database"}))

o/p:

```
{
  _id: ObjectId('69df08c95b823e6a174a704f'),
  book_id: 101,
  title: 'MongoDB Basics',
  category: 'Database',
  author: 'John Smith',
  price: 500
}
{
  _id: ObjectId('69df08c95b823e6a174a7052'),
  book_id: 104,
  title: 'SQL for Beginners',
  category: 'Database',
  author: 'David Miller',
  price: 450
}
```

---

6. Show books whose **price is greater than 600**.

db.books.find({price:{$gt:600}})

o/p:

```
{
  _id: ObjectId('69df08c95b823e6a174a7050'),
  book_id: 102,
  title: 'Python Fundamentals',
  category: 'Programming',
  author: 'Alice Brown',
  price: 650
}
{
  _id: ObjectId('69df08c95b823e6a174a7051'),
  book_id: 103,
  title: 'Data Engineering Intro',
  category: 'Data',
  author: 'Mark Lee',
  price: 800
}
{
  _id: ObjectId('69df08c95b823e6a174a7053'),
  book_id: 105,
  title: 'Machine Learning Start',
  category: 'AI',
  author: 'Sara Khan',
  price: 900
}
```

---

7. Show borrowings where **days_borrowed is greater than 5**.

db.borrowings.find({days_borrowed:{$gt:5}})

o/p:

```
{
  _id: ObjectId('69df08d55b823e6a174a7056'),
  borrow_id: 1003,
  member_id: 1,
  book_id: 103,
  days_borrowed: 7,
  borrow_date: '2026-03-03'
}
{
  _id: ObjectId('69df08d55b823e6a174a7058'),
```

```
  borrow_id: 1005,
  member_id: 5,
  book_id: 105,
  days_borrowed: 10,
  borrow_date: '2026-03-07'
}
```

Sorting

8. Display books sorted by **price descending**.

db.books.find().sort({price:-1})

o/p:

```
{
  _id: ObjectId('69df08c95b823e6a174a7053'),
  book_id: 105,
  title: 'Machine Learning Start',
  category: 'AI',
  author: 'Sara Khan',
  price: 900
}
{
  _id: ObjectId('69df08c95b823e6a174a7051'),
  book_id: 103,
  title: 'Data Engineering Intro',
  category: 'Data',
  author: 'Mark Lee',
  price: 800
}
{
  _id: ObjectId('69df08c95b823e6a174a7050'),
  book_id: 102,
  title: 'Python Fundamentals',
  category: 'Programming',
  author: 'Alice Brown',
  price: 650
}
{
  _id: ObjectId('69df08c95b823e6a174a704f'),
  book_id: 101,
  title: 'MongoDB Basics',
  category: 'Database',
  author: 'John Smith',
  price: 500
}
{
```

```
  _id: ObjectId('69df08c95b823e6a174a7052'),
  book_id: 104,
  title: 'SQL for Beginners',
  category: 'Database',
  author: 'David Miller',
  price: 450
}
```

---

9. Display members sorted by **name ascending**.

db.members.find().sort({name:1})

o/p:

```
{
  _id: ObjectId('69df08bf5b823e6a174a704a'),
  member_id: 1,
  name: 'Aarav',
  city: 'Hyderabad',
  membership_type: 'Gold'
}
{
  _id: ObjectId('69df08bf5b823e6a174a704e'),
  member_id: 5,
  name: 'Kiran',
  city: 'Hyderabad',
  membership_type: 'Gold'
}
{
  _id: ObjectId('69df08bf5b823e6a174a704b'),
  member_id: 2,
  name: 'Priya',
  city: 'Bangalore',
  membership_type: 'Silver'
}
{
  _id: ObjectId('69df08bf5b823e6a174a704c'),
  member_id: 3,
  name: 'Rahul',
  city: 'Mumbai',
  membership_type: 'Gold'
}
{
  _id: ObjectId('69df08bf5b823e6a174a704d'),
  member_id: 4,
  name: 'Sneha',
  city: 'Delhi',
```

```
  membership_type: 'Silver'
}
```

Counting

10. Count the total number of members.

db.members.countDocuments()

o/p:

5

---

11. Count the total number of books.

db.books.countDocuments()

o/p:

5

---

12. Count how many books belong to the **Database** category.

db.books.countDocuments({category:"Database"})

o/p:

2

Aggregation

13. Find the **average price of all books**.

db.books.aggregate([

  {$group:{_id:null,avg_price:{$avg:"$price"}}}

])

o/p:

```
{
  _id: null,
  avg_price: 660
```

}

---

14. Find the **maximum book price**.

```
db.books.aggregate([
  {$group:{_id:null,max_price:{$max:"$price"}}}
])
```

o/p:

```
{
 _id: null,max_price: 900}
```

---

15. Find the **minimum book price**.

```
db.books.aggregate([
  {$group:{_id:null,min_price:{$min:"$price"}}}
])
```

o/p:

```
{
 _id: null,
 min_price: 450
}
```

---

16. Find the **total days borrowed per member**.

```
db.borrowings.aggregate([
  {$group:{_id:"$member_id",total_days_borrowed:{$sum:"$days_borrowed"}}}
])
```

o/p:

```
{
 _id: 5,
 total_days_borrowed: 12
}
{
 _id: 2,
 total_days_borrowed: 3
}
{
 _id: 1,
 total_days_borrowed: 12
}
```

```
{
  _id: 3,
  total_days_borrowed: 4
}
```

---

17. Display borrowings along with **member details**.

```
db.borrowings.aggregate([
  {$lookup:{
    from:"members",
    localField:"member_id",
    foreignField:"member_id",
    as:"member_details"
  }}
])
```

o/p:

```
{
  _id: ObjectId('69df08d55b823e6a174a7054'),
  borrow_id: 1001,
  member_id: 1,
  book_id: 101,
  days_borrowed: 5,
  borrow_date: '2026-03-01',
  member_details: [
    {
      _id: ObjectId('69df08bf5b823e6a174a704a'),
      member_id: 1,
      name: 'Aarav',
      city: 'Hyderabad',
      membership_type: 'Gold'
    }
  ]
}
{
  _id: ObjectId('69df08d55b823e6a174a7055'),
  borrow_id: 1002,
  member_id: 2,
  book_id: 102,
  days_borrowed: 3,
  borrow_date: '2026-03-02',
  member_details: [
```

```
    {
      _id: ObjectId('69df08bf5b823e6a174a704b'),
      member_id: 2,
      name: 'Priya',
      city: 'Bangalore',
      membership_type: 'Silver'
    }
  ]
}
{
  _id: ObjectId('69df08d55b823e6a174a7056'),
  borrow_id: 1003,
  member_id: 1,
  book_id: 103,
  days_borrowed: 7,
  borrow_date: '2026-03-03',
  member_details: [
    {
      _id: ObjectId('69df08bf5b823e6a174a704a'),
      member_id: 1,
      name: 'Aarav',
      city: 'Hyderabad',
      membership_type: 'Gold'
    }
  ]
}
{
  _id: ObjectId('69df08d55b823e6a174a7057'),
  borrow_id: 1004,
  member_id: 3,
  book_id: 104,
  days_borrowed: 4,
  borrow_date: '2026-03-05',
  member_details: [
    {
      _id: ObjectId('69df08bf5b823e6a174a704c'),
      member_id: 3,
      name: 'Rahul',
      city: 'Mumbai',
      membership_type: 'Gold'
    }
  ]
}
{
  _id: ObjectId('69df08d55b823e6a174a7058'),
  borrow_id: 1005,
  member_id: 5,
```

```
  book_id: 105,
  days_borrowed: 10,
  borrow_date: '2026-03-07',
  member_details: [
   {
    _id: ObjectId('69df08bf5b823e6a174a704e'),
    member_id: 5,
    name: 'Kiran',
    city: 'Hyderabad',
    membership_type: 'Gold'
   }
  ]
}
{
  _id: ObjectId('69df08d55b823e6a174a7059'),
  borrow_id: 1006,
  member_id: 5,
  book_id: 101,
  days_borrowed: 2,
  borrow_date: '2026-03-08',
  member_details: [
   {
    _id: ObjectId('69df08bf5b823e6a174a704e'),
    member_id: 5,
    name: 'Kiran',
    city: 'Hyderabad',
    membership_type: 'Gold'
   }
  ]
}
```

---

18. Display borrowings along with **book details**.

```
db.borrowings.aggregate([
  {$lookup:{
   from:"books",
   localField:"book_id",
   foreignField:"book_id",
   as:"book_details"
  }}
])
```

o/p:

{
_id: ObjectId('**69df08d55b823e6a174a7054**'),
borrow_id: 1001,
member_id: 1,
book_id: 101,
days_borrowed: 5,
borrow_date: **'2026-03-01'**,
book_details: [
{
_id: ObjectId('**69df08c95b823e6a174a704f**'),
book_id: 101,
title: **'MongoDB Basics'**,
category: **'Database'**,
author: **'John Smith'**,
price: 500
}
]
}
{
_id: ObjectId('**69df08d55b823e6a174a7055**'),
borrow_id: 1002,
member_id: 2,
book_id: 102,
days_borrowed: 3,
borrow_date: **'2026-03-02'**,
book_details: [
{
_id: ObjectId('**69df08c95b823e6a174a7050**'),
book_id: 102,
title: **'Python Fundamentals'**,
category: **'Programming'**,
author: **'Alice Brown'**,
price: 650
}
]
}
{
_id: ObjectId('**69df08d55b823e6a174a7056**'),
borrow_id: 1003,
member_id: 1,
book_id: 103,
days_borrowed: 7,
borrow_date: **'2026-03-03'**,
book_details: [
{
_id: ObjectId('**69df08c95b823e6a174a7051**'),
book_id: 103,

title: **'Data Engineering Intro'**,
category: **'Data'**,
author: **'Mark Lee'**,
price: 800
}
]
}
{
_id: ObjectId(**'69df08d55b823e6a174a7057'**),
borrow_id: 1004,
member_id: 3,
book_id: 104,
days_borrowed: 4,
borrow_date: **'2026-03-05'**,
book_details: [
{
_id: ObjectId(**'69df08c95b823e6a174a7052'**),
book_id: 104,
title: **'SQL for Beginners'**,
category: **'Database'**,
author: **'David Miller'**,
price: 450
}
]
}
{
_id: ObjectId(**'69df08d55b823e6a174a7058'**),
borrow_id: 1005,
member_id: 5,
book_id: 105,
days_borrowed: 10,
borrow_date: **'2026-03-07'**,
book_details: [
{
_id: ObjectId(**'69df08c95b823e6a174a7053'**),
book_id: 105,
title: **'Machine Learning Start'**,
category: **'AI'**,
author: **'Sara Khan'**,
price: 900
}
]
}
{
_id: ObjectId(**'69df08d55b823e6a174a7059'**),
borrow_id: 1006,
member_id: 5,

```
  book_id: 101,
  days_borrowed: 2,
  borrow_date: '2026-03-08',
  book_details: [
   {
    _id: ObjectId('69df08c95b823e6a174a704f'),
    book_id: 101,
    title: 'MongoDB Basics',
    category: 'Database',
    author: 'John Smith',
    price: 500
   }
  ]
}
```

---

19. Display **member name and book title** for each borrowing.

```
db.borrowings.aggregate([
  {$lookup:{
    from:"members",
    localField:"member_id",
    foreignField:"member_id",
    as:"member"
  }},
  {$lookup:{
    from:"books",
    localField:"book_id",
    foreignField:"book_id",
    as:"book"
  }},{
  $project:{
    _id:0,
    member_name:{$arrayElemAt:["$member.name",0]},
    book_title:{$arrayElemAt:["$book.title",0]}
  }
```

```
  }
])
```

o/p:

```
{
  member_name: 'Aarav',
  book_title: 'MongoDB Basics'
}
{
  member_name: 'Priya',
  book_title: 'Python Fundamentals'
}
{
  member_name: 'Aarav',
  book_title: 'Data Engineering Intro'
}
{
  member_name: 'Rahul',
  book_title: 'SQL for Beginners'
}
{
  member_name: 'Kiran',
  book_title: 'Machine Learning Start'
}
{
  member_name: 'Kiran',
  book_title: 'MongoDB Basics'
}
```

---

20. Display **book title and total times it was borrowed**.

```
db.borrowings.aggregate([
  {$lookup:{
    from:"books",
    localField:"book_id",
    foreignField:"book_id",
    as:"book"
  }},
{ $unwind:"$book" },{
    $group:{
      _id:"$book.title",total_times_borrowed:{$sum:1}
    }
  },
  {
  $project:{
    _id:0,
    book_title:"$_id",
    total_times_borrowed:1
  }
  }
])
```

o/p:

```
{
  total_times_borrowed: 1,
  book_title: [
    'Python Fundamentals'
  ]
}
{
  total_times_borrowed: 2,
  book_title: [
    'MongoDB Basics'
  ]
}
```

```
{
 total_times_borrowed: 1,
 book_title: [
  'Data Engineering Intro'
 ]
}
{
 total_times_borrowed: 1,
 book_title: [
  'Machine Learning Start'
 ]
}
{
 total_times_borrowed: 1,
 book_title: [
  'SQL for Beginners'
 ]
}
```

---

21. Find the **total number of books borrowed by each member**.

```
db.borrowings.aggregate([
  {
    $group:{
      _id:"$member_id",
      total_books_borrowed:{ $sum:1 }
    }
  },
  {
    $lookup:{
      from:"members",
      localField:"_id",
      foreignField:"member_id",
      as:"member"
    }
  }, { $unwind:"$member" },
  {
```

```
    $project:{
      _id:0,
      member_name:"$member.name",
      total_books_borrowed:1
    }
  }
])
```

o/p:

```
{
 total_books_borrowed: 2,
 member_name: 'Kiran'
}
{
 total_books_borrowed: 1,
 member_name: 'Priya'
}
{
 total_books_borrowed: 2,
 member_name: 'Aarav'
}
```

---

22. Find the **most borrowed book**.

```
db.borrowings.aggregate([
  {
    $group:{
      _id:"$book_id",
      total_times_borrowed:{ $sum:1 }
    }
  },
  {
    $lookup:{
      from:"books",
      localField:"_id",
      foreignField:"book_id",
```

```
      as:"book"
    }
  },
  {
    $unwind:"$book"
  },
{ $sort:{ total_times_borrowed:-1 } },
  { $limit:1 },
  {
    $project:{
      _id:0,
      book_title:"$book.title",
      total_times_borrowed:1
    }
  }
])
```

o/p:

```
{
 total_times_borrowed: 2,
 most_borrowed_book: [
  'MongoDB Basics'
 ]
}
```

---

23. Find the **total borrowing count by category**.

```
db.borrowings.aggregate([
  {
    $lookup:{
      from:"books",
      localField:"book_id",
      foreignField:"book_id",
```

```
      as:"book"
    }
  }, { $unwind:"$book" },
  {
    $group:{
      _id:"$book.category",
      total_borrowings:{ $sum:1 }
    }
  },
  {
    $project:{
      _id:0,
      category:"$_id",
      total_borrowings:1
    }
  }
])
```

o/p:

```
{
 total_borrowings: 1,
 category: 'Programming'
}
{
 total_borrowings: 3,
 category: 'Database'
}
{
 total_borrowings: 1,
 category: 'Data'
}
{
 total_borrowings: 1,
 category: 'AI'
}
```

---

24. Find members who borrowed **more than one book**.

```
db.borrowings.aggregate([
  {
    $group:{
      _id:"$member_id",
      books_borrowed:{ $sum:1 }
    }
  },
  {
    $match:{
      books_borrowed:{ $gt:1 }
    }
  },
  {
    $lookup:{
      from:"members",
      localField:"_id",
      foreignField:"member_id",
      as:"member"
    }
  }, { $unwind:"$member" },
  {
    $project:{
      _id:0,
      member_name:"$member.name",
      books_borrowed:1
    }
  }
])
```

o/p:

```
{
  books_borrowed: 2,
  member_name: 'Aarav'
}
{
  books_borrowed: 2,
  member_name: 'Kiran'
}
```

Final Challenge

25. Display:

* member name

* city

* total books borrowed

Sort by **highest books borrowed first**.

```
db.borrowings.aggregate([
  {
    $group:{
      _id:"$member_id",
      total_books_borrowed:{ $sum:1 }
    }
  },
  {
    $lookup:{
      from:"members",
      localField:"_id",
      foreignField:"member_id",
      as:"member"
    }
  }, { $unwind:"$member" },
```

```
  {
    $project:{
      _id:0,
      member_name:"$member.name",
      city:"$member.city",
      total_books_borrowed:1
    }
  },
  {
    $sort:{ total_books_borrowed:-1 }
  }
])
```

o/p:

```
{
 total_books_borrowed: 2,
 member_name: 'Kiran',
 city: 'Hyderabad'
}
{
 total_books_borrowed: 2,
 member_name: 'Aarav',
 city: 'Hyderabad'
}
{
 total_books_borrowed: 1,
 member_name: 'Priya',
 city: 'Bangalore'
}
{
 total_books_borrowed: 1,
 member_name: 'Rahul',
 city: 'Mumbai'
}
```